

The Viable Selector: A Unifying Framework for Obstruction Certificates under Incomplete Observation

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Abstract

Every obstruction certificate in the calculus of viability under incomplete observation proves the same structural fact: at some information state, no single response acceptable for every compatible branch exists at the required level of the recursion. This paper develops that fact into a framework. The recursive viable-selector correspondence is defined, and the finite-horizon identity is established: an information state is viable exactly when the correspondence at it is nonempty, with the measurable-selection qualification that confines exact completeness claims to finite or suitably regular models. A meta-theorem organizes the certificate families of the companion papers — dynamic exit, admissibility, common action, tube safety, timing, certification limits, and policy-class restrictions — as emptiness conditions on correspondences obtained from one another by nesting, restriction, and label selection, with three bridging propositions making the passages precise: the correspondence ladder, timing as blind-window predecessor emptiness, and certification as label selection. The framework unifies the form of the certificates, not their cause: dynamic exit is not epistemic, the fibre criterion is a certification limit rather than a policy-viability obstruction, and the certainty-equivalence trap is a policy-class restriction that can operate under injective observation. All structural identities are verified exactly on the worked systems of the companion papers.

1. Introduction

Two companion papers develop a calculus of obstruction certificates — finite, checkable witnesses that no observation-based policy is viable. The first (Abaee, 2026a) establishes the certificate families for discrete and continuous systems; the second (Abaee, 2026b) builds the continuous-to-finite bridge for the recourse-aware mode. The mechanisms can look like a taxonomy: an exit condition here, a timing bound there, a fibre criterion, a policy-class trap. The observation of this paper is that they share one emptiness form — seven specializations of a single correspondence, not seven independent theorems, and not one cause worn seven ways either (Section 7 states the discipline). At an information state B , an observation-based regulator must choose *one* response that works for *every*

compatible branch; each certificate is a proof that, at some recursion level, no such response exists.

The paper makes this precise and then measures how much it carries.

- **The recursive viable selector** (Section 2): the correspondence $\Gamma_N(B)$ of response segments that are admissible for the declared class, implementable under the observation structure, safe through the current review interval, and recursively viable at every successor information state. Theorem 1: in finite-horizon finite models, B is viable exactly when $\Gamma_N(B) \neq \emptyset$ — and the measurable-selection qualification states exactly where pointwise nonemptiness must be upgraded to an implementable selector.
- **The meta-theorem** (Section 3, Theorem 2): declaring a policy class intersects the selector with the class's requirements; an obstruction certificate *is* a checkable condition implying the corresponding emptiness; and soundness is monotone — restricting the class or tightening the requirements preserves every verdict. The certainty-equivalence trap is the canonical instance: the unrestricted correspondence is nonempty while the class-restricted one is empty.
- **Three bridging propositions** (Section 4): the correspondence ladder $\mathcal{A}_N(B) \subseteq \mathcal{A}_{\text{tube}}(B, \Delta) \subseteq \mathcal{R}_V^B(B) \subseteq U^B(B)$, under precise hypotheses, so that emptiness at any coarser link certifies recursive nonviability; the identity of the guaranteed survival time σ^* with the horizon at which the blind-window predecessor correspondence empties, so the timing certificate *is* predecessor emptiness; and the label-selection form of the fibre criterion, $\Lambda_K(O^{-1}(y)) \neq \emptyset$ for every y , with the certainly-safe set as the maximal sound label relaxation.
- **The specializations** (Section 5): the certificate families of the companions derived as instances, in one table, with the correspondence emptied and the certificate object named in each row.
- **Verified re-derivations** (Section 6): the two-floor belief, the two-patch audit, the hidden-regime grid, and the certainty-equivalence instance of the companion papers, re-derived as framework identities in exact arithmetic (twelve check families, all passing).
- **The discipline** (Section 7): what the framework

does *not* unify. The form is common; the cause is not. Dynamic exit defeats full information and is not epistemic; the fibre criterion limits certification, not policy viability; the certainty-equivalence trap is a restriction of the policy class and can operate under injective observation.

The strongest claim the framework supports is stated at the end of Section 7 and is deliberately modest: observation-based viability requires a common recursively safe selector from information states to admissible responses; the certificates are checkable sufficient emptiness conditions; the backward recursion characterizes selector existence exactly in finite-horizon finite models; elsewhere the certificates are sound local exclusion tests.

2. The recursive viable selector

2.1 States, information, responses

Let Z be the augmented state space: a point $z \in Z$ may carry the physical state, hidden parameters, latent regimes, or institutional modes affecting admissibility. Let $\mathcal{V} \subseteq Z$ be the closed safe set. An *information state* is a nonempty set $B \subseteq Z$ of augmented states compatible with the observation record. At known times $0 = \tau_0 < \tau_1 < \dots < \tau_R$, the observation history identifies a cell of a refining partition of the compatibility relation; within a review interval I , a *response segment* α is a held action, an open-loop segment, or an observation-based feedback law, as the model admits. For a response α at B and a next observation y^+ , $\text{Succ}(B, \alpha, y^+)$ denotes the successor information state. The exposition is written for the deterministic set-valued semantics of the companions; the stochastic extension is the belief-state value recursion previewed in Section 8.

2.2 The correspondence

Fix a policy class Π , an observation structure $\mathcal{I}$, a review length Δ , and a horizon N .

Definition 1 (recursive viable selector). $\mathfrak{R}_{\Pi, \mathcal{I}}(B)$ is the set of response segments that belong to the class Π , are implementable under $\mathcal{I}$, and satisfy the nonanticipativity convention. Inductively, $\mathscr{W}_0$ is the collection of information states carried by $\mathcal{V}$, and

$$\begin{aligned} \Gamma_N(B) = \{ \alpha \in \mathfrak{R}_{\Pi, \mathcal{I}}(B) : \\ \text{every compatible trajectory remains in } \mathcal{V} \\ \text{through the interval, and} \\ \text{Succ}(B, \alpha, y^+) \in \mathscr{W}_{N-1} \\ \text{for every admissible } y^+ \}, \\ \mathscr{W}_N = \{ B : \Gamma_N(B) \neq \emptyset \}. \end{aligned}$$

Theorem 1 (finite-horizon exactness). In a finite-horizon finite model, $B \in \mathscr{W}_N$ if and only if $\Gamma_N(B) \neq \emptyset$: an N -step viable policy exists from B exactly when the recursively viable response correspondence at B is nonempty.

Proof. Backward induction on N . For $N = 0$, viability is membership in $\mathcal{V}$, which is the definition of $\mathscr{W}_0$. If B is viable for N steps, the first segment of a witnessing policy is admissible, keeps every compatible trajectory in $\mathcal{V}$ through the interval, and every possible successor information state is viable for $N - 1$ steps, so the segment lies

in $\Gamma_N(B)$; conversely, selecting recursively from nonempty Γ_k along the realized observation path assembles a viable policy. $\square$

Remark 1 (the selector qualification). Pointwise nonemptiness of $\Gamma_N(B)$ is enough for backward induction in a finite model. In general measurable spaces a policy requires an implementable *selection* $B \mapsto \alpha(B) \in \Gamma_N(B)$, and the following are distinct: nonemptiness at one specified information state; nonemptiness at every state of a collection; and the existence of a measurable nonanticipative selector on that collection. Exact completeness claims are therefore made only for finite or suitably regular models (measurable selection under standard conditions; Bertsekas and Shreve, 1978); elsewhere the certificates remain sound exclusion tests, which require no selection.

An *obstruction certificate* is any checkable condition implying $\Gamma_N(B) = \emptyset$. The certificates of the companions are all of this form; the next two sections say precisely which correspondence each empties and by which passage the emptiness is transferred to Theorem 1.

3. The meta-theorem

Declare a policy class by four requirement families: admissibility under the class rules, implementability under the observation structure, tube safety over the review interval at the declared length, and recursive viability at the declared horizon. Write

$$\begin{aligned} \Gamma(B; \Pi, \mathcal{I}, \Delta, N) = \Gamma^{\text{adm}}(B) \cap \Gamma^{\text{imp}}(B) \\ \cap \Gamma^{\text{tube}}(B; \Delta) \cap \Gamma^{\text{rec}}(B; N), \end{aligned}$$

the intersection of the correspondences enforcing each family.

Theorem 2 (meta-theorem). In a finite-horizon finite model (or a suitably regular model in the sense of Remark 1), B is viable within the declared class if and only if $\Gamma(B; \Pi, \mathcal{I}, \Delta, N) \neq \emptyset$; beyond that scope the identity holds in the selector-qualified sense of Theorem 1 (the nonemptiness-to-viability direction with an implementable selection). Consequently: (i) a checkable condition implying the emptiness of any one factor is an obstruction certificate for the whole class; (ii) the certificate semantics of the companions — a finite, policy-independent list of adversarial tests whose violation is forced — are exactly the dual witnesses of such emptinesses; (iii) soundness is monotone: if $\Pi' \subseteq \Pi$, or if any factor correspondence is tightened, then $\Gamma(B; \Pi', \mathcal{I}, \Delta, N) \subseteq \Gamma(B; \Pi, \mathcal{I}, \Delta, N)$, so every emptiness verdict for the larger class remains valid for the smaller, and new verdicts may appear.

Proof. (i) and the identity are Theorem 1 with the class requirements carried in $\mathfrak{R}_{\Pi, \mathcal{I}}$; (ii) is the certificate semantics of the companions restated (a Farkas vector, an adversarial label list, or a timing bound is a finitely checkable witness of the emptiness of the relevant factor); (iii) is immediate from the intersection form. $\square$

Remark 2 (the three selector objects and the nesting). Three objects share the notation and must be kept distinct. $\Gamma_N(B)$ of Definition 1 is the inductive correspondence. The four-factor object $\Gamma(B; \Pi, \mathcal{I}, \Delta, N)$ is its *one-step unpacking*: the recursive factor $\Gamma^{\text{rec}}(B; N)$ is the inductive one, requiring $\text{Succ}(B, \alpha, y^+) \in \mathscr{W}_{N-1}$ — so the intersection is inductive, not a flat static product. $\mathcal{A}_N(B)$ of Proposition 1 is the singleton-action reading of the same object. Moreover, under the standing hypotheses the factors are *nested*, $\Gamma^{\text{rec}}(B; N) \subseteq \Gamma^{\text{tube}}(B; \Delta) \subseteq \Gamma^{\text{safe}}(B) \subseteq \Gamma^{\text{adm}}(B)$ (Proposition 1), so the

four-fold intersection coincides with its finest factor: the factorization does not assert independent constraints — it *locates* each certificate family at a factor and carries the monotonicity of part (iii). Emptiness at any factor is thus emptiness along the filtration, and the coarser the factor, the more easily the emptiness is certified.

Corollary 1 (policy-class obstructions). If $\Gamma(B; \Pi, \mathcal{I}, \Delta, N) \neq \emptyset$ but $\Gamma(B; \Pi', \mathcal{I}, \Delta, N) = \emptyset$ for a restriction $\Pi' \subset \Pi$, then B is viable under Π and nonviable under Π' . In the certainty-equivalence trap of the companion paper, Π' is the declared class of laws that act on the uncorrected reading $\hat{z}$ as if it were the state (the canonical such law: hold the reading constant); the restriction removes the required selector even under an injective observation. The scope alignment with the companion paper is exact: its remark is policy-specific for its own instance, and the class form here is carried by the verified per-law drift bound of Section 6 (canonical-law form) — the full-class quantifier over every uncorrected law is the successor architecture’s layer-five reading and is cited, not re-claimed.

Table 1 instantiates the theorem: each certificate family of the companions is the emptiness of one factor, or of a correspondence derived from one by the passages of Section 4.

4. Bridging propositions

Proposition 1 (the correspondence ladder). Let B be an information state and $\Delta \leq$ one review interval. Under the standing hypotheses of the companions (Abae 2026a, Section 4: constraint functions of class C^1 ; disturbance correspondences with closed graph; dynamics admitting the comparison argument), the recursively viable, tube-safe, instantaneously safe, and admissible correspondences satisfy

$$\mathcal{A}_N(B) \subseteq \mathcal{A}_{\text{tube}}(B, \Delta) \subseteq \mathcal{R}_V^B(B) \subseteq U^B(B),$$

where $\mathcal{A}_N(B) = \Gamma_N(B)$ with the singleton action reading. Each inclusion can be strict. Consequently every emptiness certificate at a coarser link is a certificate of recursive nonviability at the finer one.

Proof. An action viable for N steps keeps every compatible trajectory in $\mathcal{V}$ over the first interval, hence is tube-safe at Δ ; a tube-safe action is, at every active boundary state of every compatible branch, a direction of nonnegative constraint drift against every admissible disturbance (otherwise a measurable adverse selection exits within the interval), hence instantaneously safe; instantaneous safety requires admissibility at every branch. Strictness is real: the two-patch instance of Section 6 has $\mathcal{A}_{\text{tube}}(\cdot, 2)$ empty where $\mathcal{R}_V^B$ is a singleton. $\square$

Proposition 2 (timing as blind-window predecessor emptiness). For the delayed-information model (no information before T_{obs} , revelation at T_{obs}), let $\text{Pre}_{\text{blind}}(\mathcal{C})$ be the collection of information states admitting a blind response — admissible for every branch, common across indistinguishable branches, within the declared window class Π_{win} — that keeps every branch in $\mathcal{V}$ through the window with every revealed successor in $\mathcal{C}$. Write $\text{Pre}_{\text{blind}}^t(\mathcal{C})$ for the t -fold iteration of the predecessor map, and say that B_0 *survives at horizon t* (within the class) when $B_0 \in \text{Pre}_{\text{blind}}^t(\mathcal{C})$ for the revealed-successor collections $\mathcal{C}'$ along the way. Let $\sigma_{\Pi_{\text{win}}}^*(B_0)$ be the guaranteed blind-window survival time of the declared class — the class index is load-bearing, since different classes empty at different horizons. Then

$$\begin{aligned} \sigma_{\Pi}^*(B_0) &= \sup\{t : B_0 \text{ survives at horizon } t \text{ under } \text{Pre}_{\text{blind}}^{\Pi}\}, \\ \sigma_{\Pi}^*(B_0) < T_{\text{obs}} &\implies B_0 \notin \text{Pre}_{\text{blind}}^{\Pi, T_{\text{obs}}}(\mathcal{C}). \end{aligned}$$

The timing certificate of the companions is exactly this emptiness, evaluated at the observation horizon; and the declaration

of the window class is load-bearing, with exact boundary values on the audited grid (Section 6, hidden-regime grid): the declared hold class survives to horizon k exactly when $z_0 \geq 1 + k$ (discrete survival $\sup \lfloor z_0 - 1 \rfloor$, matching the successor’s continuous $\sigma^*(z_0) = z_0 - 1$ level sets), while the unrestricted sequential class survives exactly when $z_0 \geq 2$ for $k \geq 2$.

Proof. The sup-identity is the definition of σ^* read through Theorem 1: survival to horizon t is precisely nonemptiness of the blind predecessor correspondence at t . If $\sigma^*(B_0) < T_{\text{obs}}$, no blind response keeps every branch safe to the horizon, so the correspondence at T_{obs} is empty whatever post-reveal collection $\mathcal{C}$ is offered. $\square$

Proposition 3 (certification as label selection). For a label space $\{0, 1\}$ and the safety label $\mathbf{1}_K$, define the exact-label correspondence on a set F of augmented states by $\Lambda_K(F) = \{\ell \in \{0, 1\} : \ell = \mathbf{1}_K(z) \ \forall z \in F\}$. An exact observation-only certifier exists if and only if $\Lambda_K(O^{-1}(y)) \neq \emptyset$ for every observation y ; the certainly-safe set is $\{y : \Lambda_K(O^{-1}(y)) = \{1\}\}$, the maximal sound label relaxation. The fibre criterion is the label-selection analogue of common-action selection at the zeroth level of the recursion; it is a limit on certification, not a policy-viability obstruction (Section 7).

Proof. A certifier is a map $\ell : O(Z) \rightarrow \{0, 1\}$ with $\ell(y) = \mathbf{1}_K(z)$ for all $z \in O^{-1}(y)$; such a value exists exactly when $\mathbf{1}_K$ is constant on the fibre, which is nonemptiness of $\Lambda_K(O^{-1}(y))$. The certainly-safe set collects the fibres on which the only constant label is the safe one. $\square$

5. The specializations

The four one-step mechanisms of the companion paper are the static intersection object at four choices of acceptable response, evaluated at the information state — the mechanism is the intersection across compatible branches, $\Gamma^{\text{adm}}(B) = \bigcap_{z \in B} U(z)$ for admissibility, $\Gamma^{\text{safe}}(B) = \bigcap_{z \in B} \mathcal{R}_V(z)$ for instantaneous safety, $\Gamma^{\text{tube}}(B; \Delta) = \bigcap_{z \in B} \mathcal{A}_{\text{tube}}(z, \Delta)$ for tube safety at a review length (the held action reading) — and the epistemic-emptiness construction, which is the admissibility register evaluated on a merged observation fibre — one constant observation, two admissible-response sets with empty intersection. The dynamic exit certificate is the $N = 0$, singleton-information-state base case of the same emptiness form: at a single branch the intersection is trivial, and the drift condition empties the boundary response set outright — no information-theoretic content, and no recursion needed. The delayed-information obstruction is Proposition 2: the same intersection, demanded through the blind window and handed to the revealed successors. The certification limit is Proposition 3. The certainty-equivalence trap is Corollary 1. One object — a response acceptable for every compatible branch at the required recursion level — seven emptinesses.

6. Verified re-derivations

The framework’s identities are machine-checked in exact rational arithmetic (`paper2_viable_selector_v3_verify.py`, standard library; twelve check families, all passing) on the worked systems of the companions. *Static intersection (two-floor belief)*: the common admissible correspondence is $[0, 1]$ — nonempty — while the common safe correspondence is empty: the common-action certificate is exactly $\Gamma^B = \emptyset$. *Label selection (two-floor index and its two-dimensional aggregate)*: the crossing fibre has $\Lambda_K = \emptyset$, the safe fibre $\Lambda_K = \{1\}$, and the certainly-safe readings are exactly the aggregate values $I \geq \frac{7}{5}$ (seven grid cells), matching the companions’ computation. *Ladder (two-patch system of the calculus paper’s protection audit — patches $i \in \{1, 2\}$ with stocks $z_i \in \{0, 1, 2, 3\}$,*

certificate family	correspondence emptied	certificate object	source
dynamic exit (base)	the $N = 0$ singleton-information-state base case of the emptiness form: the boundary response set at full information, emptied outright by the drift condition	drift/comparison bound, exit time $\leq q(x_0)/\varepsilon$	Abae 2026a, Thm. (exit)
admissibility	$\Gamma^{\text{adm}}(B) = \bigcap_z U(z)$	empty common admissible set	Abae 2026a (admissibility register)
common action	$\Gamma^{\text{safe}}(B) = \bigcap_z \mathcal{R}_V(z)$	empty common safe-action set; Farkas or $m+1$ witnesses	Abae 2026a, Thm. (common action)
tube safety	$\Gamma^{\text{tube}}(B; \Delta)$	empty tube-safe set at review length Δ	Abae 2026a (ladder)
delayed information (timing)	blind-window predecessor correspondence at horizon T_{obs} (Prop. 2)	$\sigma^*(B_0) < T_{\text{obs}}$; LP values and Farkas duals	Abae 2026a (delayed); Abae 2026b (finite LP)
certification limit policy class	label selector $\Lambda_K(O^{-1}(y))$ (Prop. 3) $\Gamma(B; \Pi', \cdot)$ for the restricted class (Cor. 1)	a fibre across the safe-set boundary emptiness under the restriction	Abae 2026a (fibre criterion) Abae 2026a (certainty-equivalence trap)

Table 1: The certificate families of the companion papers as emptiness conditions on declared-class correspondences. The last three rows act on correspondences derived by the passages of Propositions 2 and 3 and Corollary 1; the fibre row is a certification limit, not a policy-viability obstruction (Section 7).

recruitment +1 per step cap 3, unprotected-patch damage 2, instruments $u \in \{1, 2\}$ protecting one patch, floors $z_i \geq 1$; defined in Abae 2026a, Section 8, and restated with full data in the verification script): $\mathcal{A}_{\text{tube}}(\cdot, 2) \subseteq \mathcal{R}_V^B \subseteq U^B$ holds at every state, is strict at exactly five — at $(1, 2)$ the two-step held-action correspondence is empty while the instantaneous one is $\{1\}$ — and tight at exactly four (the counts are asserted exactly by the record). *Recursive identity (two-patch audit):* the Γ_N correspondence recursion and the belief-kernel recursion return identical verdicts on all six audited beliefs (three viable, with witness selectors; three nonviable, each certified by common-action emptiness), and singleton Γ_8 nonemptiness coincides with the full-information kernel. *Timing (hidden-regime grid, all 48 cells):* the blind-window predecessor correspondence over the declared hold class is nonempty at horizon k exactly when $z_0 \geq 1+k$ — the viability boundary — while the unrestricted sequential class is nonempty exactly when $z_0 \geq 2$ for $k \geq 2$: Proposition 2’s load-bearing declaration, verified. *Policy class (the certainty-equivalence instance, canonical-law form):* the corrected law has drift identically zero on $[1, 2]$, so the unrestricted correspondence is nonempty, while the canonical certainty-equivalence law — the one acting on the uncorrected reading — has drift at least $\frac{21}{100} > 0$ there and exits the band in finite time (direct iteration): the correspondence restricted to that law is empty, Corollary 1 attained on the canonical restriction. The quantifier over the whole uncorrected class is the successor architecture’s layer-five reading; this paper verifies and claims the canonical-law instance and cites the rest. *Factor identity (audit addition):* on the six audited beliefs the four factor correspondences nest exactly as Proposition 1 states, and the four-fold intersection equals its finest factor — the meta-theorem’s factorization locates certificates along a filtration (checks V7–V11).

7. What is not unified

The framework unifies the *form* of the certificates — emptiness of a response correspondence at an information state — not their *cause*, and conflating the two would misdescribe the companions. The dynamic exit certificate is not epistemic: it defeats full-information control. The fibre criterion is not a policy-viability obstruction: it concerns exact certification, a verdict rather than a control. The certainty-equivalence trap is not necessarily informational: it can occur under an injective observation, because the restriction is on the policy class. Accordingly, the strongest claim the framework supports is:

observation-based viability requires the existence of a common recursively safe selector from information states to admissible responses; the obstruction certificates of the companions are checkable sufficient conditions under which no such selector exists; in finite-horizon finite systems the backward recursion characterizes selector existence exactly; in continuous or infinite-horizon settings the certificates are sound local exclusion tests. Within those limits the selector framework is the spine on which the companion results, and the extensions that follow, assemble.

8. Outlook: the stochastic selector

The belief-state safety value of the companion paper’s probabilistic development is the stochastic extension of the recursive selector: Γ_N replaced by the value recursion over distributional information states, emptiness replaced by deficit from unity. Its exact rational theory — piecewise-linear values, support degeneration, and the agreement between certificate boundaries and value boundaries on the audited classes — is developed in the sequel, whose second edition also quantifies the class declaration used in Proposition 2.

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Declarations

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